

M82 Cigar Galaxy — UQFF Starburst Superwind

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PAPER_784: M82 Cigar Galaxy — UQFF Starburst Superwind

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Abstract

M82 (NGC 3034), the “Cigar Galaxy,” is the archetypal starburst galaxy, located only ~12 million light-years away ($z \approx 0.0008$) in Ursa Major. Tidally disturbed by its companion M81, M82 experiences a star-formation rate roughly $10\times$ higher than the Milky Way, driving a spectacular bi-polar superwind of hot gas and dust erupting ~12 kly above and below the disk. The superwind magnetic field reaches $\sim 10^{-4}$ T — characteristic of the starburst regime. Under UQFF, $v = 106$ m/s (superwind velocity) and $B = 10^{-4}$ T (starburst-amplified field) yield $g_{M82} \approx 1.053 \times 10^{-1}$ m/s², identical to the Tarantula Nebula and Stephan’s Quintet at these extreme parameters.

1. Introduction

M82’s starburst was triggered ~100 Myr ago by a close encounter with M81. The resulting disk starburst currently produces ~10 $M_{\odot}$ /yr in a region only ~1 kpc in diameter — one of the most concentrated starbursts in the nearby universe. The galactic-scale superwind reaches ~1,000 km/s and carries a luminosity of ~ 10^{41} erg/s. Radio measurements confirm B-fields of ~50–200 μ T throughout the starburst disk. UQFF encodes the superwind through $v = 106$ m/s and the starburst-amplified $B = 10^{-4}$ T, placing M82 in the UQFF starburst regime alongside Tarantula 30 Dor (PAPER_774) and Stephan’s Quintet (PAPER_778).

2. Master UQFF Gravity Equation

$$g_{M82}(r, t) = (G \times M)/r^2 \times (1 + H(z) \times t) \times (1 + M_s f) \times (1 + f_T RZ) + a_E M$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	1010 MM_sun = 1.989×1040 kg	NED
Disk radius	r	2×1020 m (~21 kly)	NED
SFR	—	10 MM_sun/yr	Radio/IR
Age	t	1×108 yr = 3.156×1015 s	Starburst duration
M_sf	—	0.15	UQFF starburst mass fraction
Redshift	z	0.0008	Spectroscopic
v_EM	v	106 m/s	Superwind velocity
B_EM	B	10-4 T	Starburst-amplified B
f_TRZ	—	0.05	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$g_{grav} = 6.6743e-11 \times 1.989e40 / (2e20)^2 = 3.319e-11 m/s^2$$

Step 2: Cosmic Expansion Factor

$$H(z) = 2.268e-18 s^{-1}; H(z) \times t = 2.268e-18 \times 3.156e15 = 7.160e-3; factor = 1.00716$$

Step 3: SFR Mass Fraction (Starburst)

$$M_s f = 0.15; 1 + M_s f = 1.15$$

Step 4: Time-Reversal Correction

$$f_T RZ = 0.05; 1 + f_T RZ = 1.05$$

Step 5: Gravitational Total

$$g_{grav_total} = 3.319e-11 \times 1.00716 \times 1.15 \times 1.05 = 4.015e-11 m/s^2$$

Step 6: Aether EM Correction (Starburst Level)

$$v = 106 m/s, B = 10^{-4} T$$

$$a_E M = (1.602e-19 \times 106 \times 10^{-4} / 1.673e-27) \times 11 \times 10^{-12} = 1.053e-1 m/s^2$$

Step 7: Final Solution

$$g_M^{82} = 4.015e-11 + 1.053e-1 \approx 1.053e-1 m/s^2$$

4. Physical Interpretation

At only 12 Mly distance, M82 is the closest prototype of the starburst superwind regime. The observed superwind velocity $\sim 1,000$ km/s and starburst B-field $\sim 10^{-4}$ T both directly confirm the UQFF starburst parameters. M82's result $g = 1.053 \times 10^{-1}$ m/s² matches Tarantula Nebula (PAPER_774) and Stephan's Quintet (PAPER_778), confirming UQFF universality across: dwarf-scale starburst (30 Dor in LMC), compact-group intergalactic shock (Stephan's Quintet), and galaxy-scale starburst (M82) at the same extreme EM parameter combination ($v = 106$ m/s, $B = 10^{-4}$ T).

5. Conclusions

UQFF applied to M82 yields $g \approx 1.053 \times 10^{-1}$ m/s², confirming M82 occupies the same UQFF starburst-shock class as Tarantula and Stephan's Quintet. At $z = 0.0008$, the nearest starburst galaxy serves as the closest-distance validation point for the UQFF starburst regime.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **galaxy-rotation** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation`

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{rot}})(\partial^\mu \phi_{\text{rot}}) - V(\phi_{\text{rot}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{rot}}) = \frac{1}{2}m^2\phi_{\text{rot}}^2 + \frac{\lambda}{4!}\phi_{\text{rot}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{rot}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{rot}}} = v_c^2/r - GM/r^2 - F_{U_Bi_i}/(m \cdot r) + \rho_{\text{vac},[\text{SCm}]} \cdot \Omega^2 r = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{rot}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.112 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 11, \quad n_{\text{channel}} = 5/26$$

Since $p_{\text{DVP}} = 11$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **109 yr** (disk settling timescale):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.112	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 11$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	$F_{\text{NeutronForce}}$, σ_{SCm} , SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \cdot \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}} \cdot (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 \cdot \Gamma_2)}] \cdot (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \cdot \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_theta_q26.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a; q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n\}2} / (-q; q)_n$ - $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n\}2} / (-q^{2;q}2)_n$ - $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n\}2} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \text{Prod}_{\{i=1\}^{26}} [1 + [SSq]\exp(-kappa_i*n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).